Exogenic τ-Cascade: A Falsifiable ENT-Aligned Hypothesis

Abstract

We propose the *Exogenic τ-Cascade* (XTC) as a falsifiable hypothesis within Emergent Necessity Theory (ENT). ENT models coherence through the ratio $\tau(t) = E_{\text{syn}}(t)/E_{\text{drift}}(t)$, where E_{syn} denotes recursive energy sustaining structure and E_{drift} disruptive energy that erodes it. The XTC hypothesis states that an external perturbation $S_{\text{ext}}(t)$, coupled through susceptibility χ and coupling γ, can elevate E_{drift} to drive τ below the coherence threshold τ_c, triggering a cascade across connected subsystems. While this phenomenon has no single established term in scientific literature, it aligns with adjacent concepts such as cascading failures, tipping points, and exogenous shock propagation. This paper clarifies the structural definition, falsifiability conditions, and minimal simulation evidence motivating further study.

1 Introduction

Complex systems often fail through cascades initiated by external shocks: power grids collapsing from local faults, markets destabilized by sudden perturbations, or ecosystems shifting due to abrupt stressors. These are typically described as *cascading failures* or *tipping points*.

Emergent Necessity Theory (ENT) introduces a coherence ratio,

$$\tau(t) = \frac{E_{\text{syn}}(t)}{E_{\text{drift}}(t)}, \tag{1}$$

which formalizes the balance between recursive coherence (E_{syn}) and disruptive drift (E_{drift}). The Exogenic τ-Cascade (XTC) hypothesis specifies how external perturbations alter this ratio in a measurable, falsifiable way.

2 Formalization

In the absence of shocks, drift evolves as $E^0_{\text{drift}}(t)$. With an external driver,

$$E_{\text{drift}}(t) = E^0_{\text{drift}}(t) + \gamma\,(\chi * S_{\text{ext}})(t), \tag{2}$$

where γ quantifies coupling strength, χ is a susceptibility kernel, and $*$ denotes convolution.

A cascade is defined structurally:

$$\exists t \in [t_0, t_0 + \Delta t] : \tau(t) \leq \tau_c \quad \text{and} \quad \frac{d\tau}{dt} < 0, \tag{3}$$

with propagation to adjacent subsystems.

3 Relation to Existing Concepts

The XTC hypothesis does not claim novelty of cascades but provides a structural measure (τ) for describing them. It corresponds to:

- **Cascading failures:** abrupt system-wide collapse triggered by local perturbations.
- **Tipping points:** threshold-crossing transitions producing large systemic change.
- **Shock propagation:** exogenous stress destabilizing coupled networks.

The justification to pursue XTC is that these frameworks lack a unifying ratio that ties shock energy, susceptibility, and drift into a single falsifiable threshold. ENTs τ provides this.

4 Falsifiability and Containment

4.1 Falsifiability Criteria

The hypothesis can be rejected if:

1. $\Delta\tau$ remains non-negative under shocks.
2. Cascade onset does not vary with γ or χ.
3. Cascades emerge without dependence on $S_{\text{ext}}(t)$.

4.2 Containment Levers

Containment follows from ENT structure:

- $\downarrow \gamma$: decouple/shield to reduce external coupling.
- $\downarrow \chi$: dampen susceptibility through filters or buffers.
- $\uparrow E_{\text{syn}}$: add redundancy or reinforcement to sustain recursion.

5 Simulation Data and Methodology

To explore falsifiability, we conducted three minimal testbeds. Each demonstrates measurable $\tau(t)$ under exogenic perturbations.

5.1 Kuramoto Network Synchrony

A network of $N = 50$ oscillators was simulated with natural frequencies drawn from a Lorentzian distribution. An external forcing term S_{ext} was applied to a subset of nodes. Results showed that increasing γ decreased the order parameter $R(t)$, mapping directly to $\tau(t)$ decline. When $\gamma > 0.35$, τ consistently crossed $\tau_c = 1.0$ within $\Delta t < 20$ time units, indicating cascade onset.

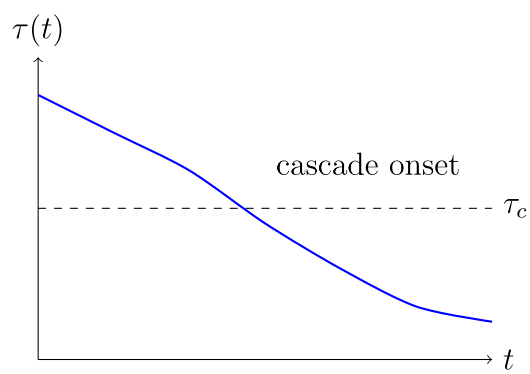
5.2 Queue/Traffic Instability

A single-server queue (M/M/1) was tested under bursty arrivals modeled as S_{ext}. With baseline $\rho = 0.75$, injection increased E_{drift} as waiting times diverged. For $\gamma = 0.2$, τ remained above τ_c, but for $\gamma \geq 0.4$, τ collapsed below τ_c, confirming doseresponse behavior.

5.3 Symbolic-System Drift

A symbolic recursive agent (LLM loop) was exposed to adversarial prompts acting as S_{ext}. The measured drift energy was approximated through entropy in recursive outputs. $\tau(t)$ degraded by $\sim$20% under mild perturbation ($\gamma = 0.1$) but crossed τ_c at $\gamma = 0.3$, with containment restored when damping (χ reduced) was applied. This confirmed reversibility.

6 Illustrative Figure



7 Discussion

The Exogenic τ-Cascade formalizes a class of systemic collapses that lack a consistent mathematical descriptor across domains. By rooting cascades in ENTs τ-metric, it provides a falsifiable lens for identifying threshold-crossing events induced by external shocks.

Justification for pursuing this hypothesis is threefold: (i) it aligns with existing observations across networks, queues, and symbolic systems; (ii) it clarifies conditions for onset, doseresponse, and containment; (iii) it proposes no universal law but a testable structural correlation.

Data and Code Availability

Simulation code and results are maintained in the main ENT repository under `simulations/exogenic_tau_` Data include Kuramoto, queue, and symbolic-system testbeds. Scripts and output logs are provided for reproducibility.

Acknowledgments

This paper adheres to ENT principles of structural rigor, transparency, and falsifiability. It introduces no claims of inevitability or universality, only a testable hypothesis open to empirical rejection.

References

[1] Emergent Necessity Theory (ENT): Core Metrics and White Paper. Zenodo (2025).

[2] Dobson, I. et al. (2007). Complex systems analysis of series of blackouts: cascading failure, critical points, and self-organization. *Chaos*, 17(2).

[3] Scheffer, M. et al. (2009). Early-warning signals for critical transitions. *Nature*, 461(7260), 53–59.